# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**First Semester of M. Pharm (Clinical Pharmacy)**

**University Theory Examination December-2016**

**PHCC004 Analytical Techniques**

**Date: 12/12/2016, Monday** **Time:10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

**Instructions:**

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

## SECTION-I

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|---|---|---|---|
| **Q.1** | **(A)** | Give the applications of Spectroflourimetry. | 5 |
| | **(B)** | Explain fluorescence and phosphorescence. Discuss advantages and limitations of Spectrofluorimetry in comparison with UV visible spectroscopy. | 5 |
| **Q.2** | **(A)** | Discuss sample preparation and principle of immune-electron microscopy. | 5 |
| | **(B)** | Classify the techniques of *in vivo* cell imaging and explain any one in detail giving suitable example. | 5 |
| | | **OR** | |
| **Q.2** | **(A)** | What is flow cytometry? Explain its working principle and instrumentation. | 5 |
| | **(B)** | Write the principle of immune-flouroscence Microscopy and classify the techniques giving suitable example. | 5 |
| **Q.3** | | **Answer the following (Any three)** | 15 |
| | **(A)** | Give the applications of ELISA. | |
| | **(B)** | Enlist techniques used for live cell imaging and explain phase contrast microscopy giving suitable example. | |
| | **(C)** | Write the principle, applications and protocol of lymphocyte proliferation. | |
| | **(D)** | Discuss zone electrophoresis. | |

## SECTION-II

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|---|---|---|---|
| **Q.4** | **(A)** | Discuss factors affecting absorption position ($\lambda$max) in UV spectroscopy giving suitable examples. | 5 |
| | **(B)** | Write about sample handling techniques in IR spectroscopy. | 5 |
| **Q.5** | | Explain derivatization in HPLC and HPTLC. | 10 |
| | | **OR** | |
| **Q.5** | | Classify chromatographic technique. Discuss the principle and instrumentation for flash chromatography. | 10 |
| **Q.6** | | **Answer the following (Any three)** | 15 |
| | **(A)** | Explain Factor affecting Chemical Shift in detailed. | |
| | **(B)** | Explain principle of Mass spectroscopy. Classify ionizing techniques used in Mass spectroscopy. Describe any one in detail. | |
| | **(C)** | Mc Lafferty and Diel's Alder rearrangement. | |
| | **(D)** | Explain instrumentation for NMR. | |

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